Chan 2021 Small-Cell Lung Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

24/34 samples (≥37 cells; dominant 92→15 spillover cliff) intermix on one continuous manifold — no per-patient islands (radius-graph: 1 detached cell; iLISI median 1.00→8.79 of 24); both cell-cycle compartments (Body 38,882 / Cycling 12,116) carry multiple samples.

Gate 2 — module geography → GO *(non-branching)*

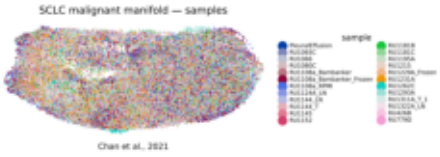
Single embryonic axis, no hub-into-arms. Gast & pSom co-vary (smoothed r=+0.28; within-patient median r=+0.06, 15/23 samples positive; joint-high enrichment 1.42) — overlap, not competition. Crest is an inverse terminal gradient (pSom↔Crest r=−0.89; joint pSom&Crest-high enrichment 0.20). No Crest-on-Gast-arm inversion to flag (there are no arms).

Gate 3 — hub mutual exclusivity → NA

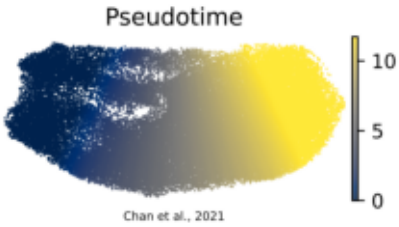
Non-branching: the PPT tree over-forks the compact cloud (9–17 spurious forks across the grid, no single dominant Y) and Gast/pSom co-vary rather than compete — there is no hub or branch pair to test.

STEP FIGURES

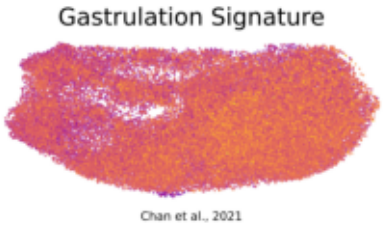
Gate 1: integration



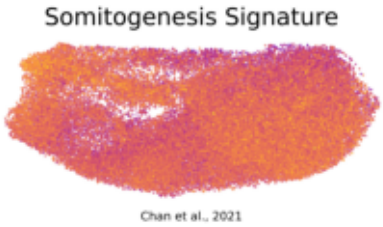
Trajectory / root



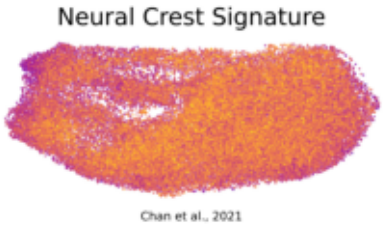
Gast



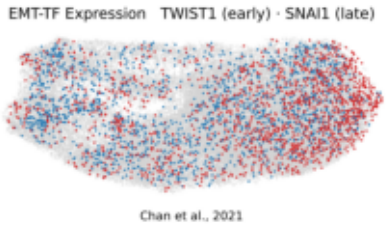
pSom



Crest



EMT-TFs



Dataset. SCLC (Chan et al., 2021), 10x — 34 patients / 51,052 malignant cells (86,662 atlas). Threshold ≥37 keeps 24 samples (50,999 cells); the whole malignant manifold (50,998 after island removal, no lobe scoping) enters the trajectory. Reported subtypes: SCLC-A 29,099 / SCLC-N 18,176 / SCLC-P 2,989 / NSCLC 788.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The SCLC malignant manifold is a single continuous axis: Gast and pSom overlap (co-vary within and across patients) while Crest is the inverse/terminal readout — the embryonic programmes are present but not resolved into two competing arms, so there is no Puram-style hub-and-spoke bifurcation and no hub to test. Subtype is fully patient-confounded (24/24 samples >80% one SCLC subtype), so Harmony-on-sample intermixes SCLC-A/N as expected — not an erased bifurcation. Modules are cell-cycle-free (|r| with cycling <0.14) and the pSom↔Crest axis strengthens in non-cycling cells. Caveat: this near-isotropic cloud (axis ratio ~1.6) has no strong 1-D trajectory — the curve tracks the dominant Gast/proliferation long-axis while the pSom→Crest gradient runs oblique, so Gate-2 geography is read off the signature embeddings, not the pseudotime ordering (Crest ρ vs pt only +0.07). EMT-TFs (widened panel): TWIST1 (early/root) → SNAI1 (late/terminus); ZEB1 is the most-expressed factor but pseudotime-flat, so not an endpoint marker.

Parameter choices

Min cells/sample	≥37 (24/34; dominant 92→15 cliff)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**n_pcs**	20 (swept 20/40; robust range)
Geometry	Linear (non-branching)	**EMT-TFs**	TWIST1 (early) → SNAI1 (late)